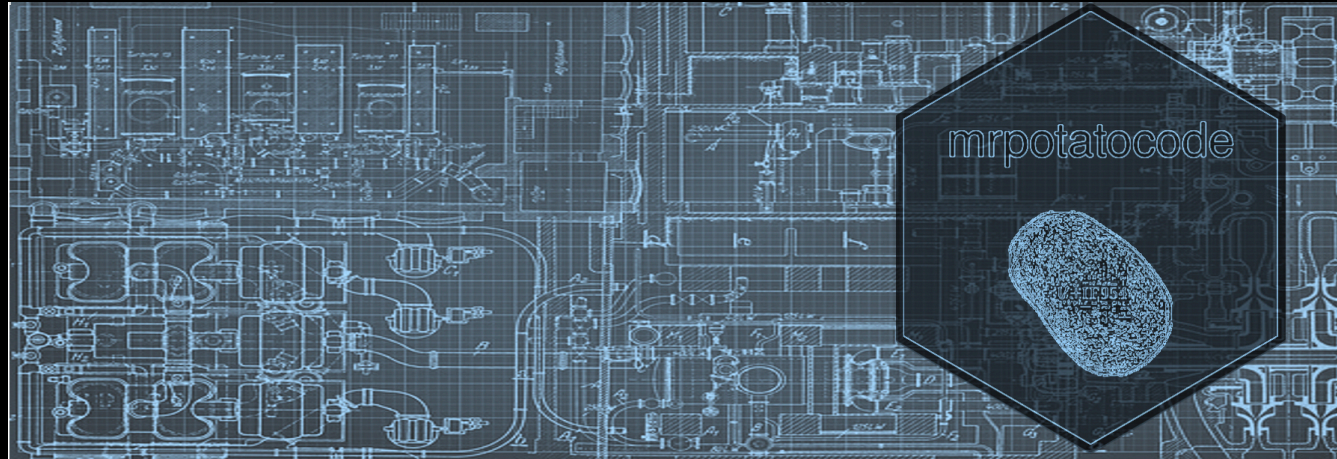




Introduction to Data Access and Storage

Thomas Rosenthal – DSI @ UofT

Session 02

Building Queries:

Fundamental Three Commands

Two More Commands

Putting Things Together with JOIN

Building Queries:

Fundamental Three Commands

Two More Commands

Putting Things Together with JOIN

Fundamental Three Commands

SELECT

FROM

WHERE

Fundamental Three Commands

- Our first three commands (`SELECT`, `FROM`, `WHERE`) are essential to nearly every SQL query
- The template for our initial SQL statement is as such:

`SELECT` : *the columns we want to retrieve*

`FROM` : *the table we are querying*

`WHERE` : *filters/conditions (optional)*

`ORDER BY` : *column sorting: ascending `ASC` or descending `DESC` (optional)*

`LIMIT` : *how many rows we want to return (optional)*

Fundamental Three Commands

- Always specified in this order:
 - `SELECT` will come first
 - `FROM` will come after `SELECT`
 - when we are querying more than one table at a time, each will come after `FROM` but before `WHERE` (more on this later)
 - `WHERE` will come after `FROM`
 - `ORDER BY` will come after `WHERE` clauses
- We'll sometimes use the `LIMIT` clause to look at data
 - This comes at the very end of a query
 - `LIMIT` shouldn't be used for analytics unless you have a specific reason
 - `ORDER BY` often impacts the usefulness of `LIMIT`
- Remember:
 - In SQL, we use two dashes `--` to comment out lines, rather than `#`

Fundamental Three Commands

SELECT

FROM

WHERE

SELECT

- At its simplest `SELECT` specifies column names we are retrieving
 - commas come between each column name
 - `SELECT student, course, grade ...`
 - column names with a space need to be enclosed in square brackets
 - `SELECT [poorly named column], better_column_name, AnotherColumnName`
- Within `SELECT` statements we can perform manipulations on columns
 - e.g. rename a column
 - `SELECT [poorly named column] AS better_col`
 - combine two text columns
 - perform math on a numeric column
 - ...and many more things

SELECT

- We can use `SELECT` to perform math without a `FROM` statement
 - `SELECT 1 + 1`
 - `SELECT 10*5, cos(2), pi()`
- And we can use `SELECT` to specify constant values
 - `SELECT 2026 AS this_year, 'March' AS this_month`
- When selecting columns, they need to exist in the table!

Fundamental Three Commands

SELECT

FROM

WHERE

FROM

- `FROM` statements indicate which table the data is from and where the table is located
 - in more complicated RDBMs, you will often have multiple databases on the same server and multiple schema within those databases
 - a fully qualified location of a table would thus be `database.schema.table`
- `SELECT * FROM table_name` indicates *everything* in the table
- Best practice suggests that we should explicitly call each column, even if we want all of them
 - **Why do we think this is the case?** 🗨️💬 **Think, Pair, Share**

FROM

(SELECT & FROM live coding)

SQL programmers be like



Image: Unknown meme creator

Fundamental Three Commands

SELECT

FROM

WHERE

WHERE

- **WHERE** clauses are conditions that the query will follow
- When we want to have multiple conditions, we use a single **WHERE** and then additional logical operations

```
SELECT *  
FROM students  
WHERE first_name = 'Thomas'  
AND last_name = 'Rosenthal'
```

- **Notice we put string values in single quotes**
 - SQLite also allows double quotes, with a few minor caveats
- **WHERE** clauses always return rows evaluating to **TRUE**
 - Follows Boolean rules if more than one condition is present

WHERE

Logical Operators

- AND
- OR
- NOT
- NOT IN
- equals: =
- does not equal: <> !=
 - (flavour dependent)
- greater than (equal to): > >=
- less than (equal to): < <=
- BETWEEN
- EXISTS
 - table specific
- IS
 - NULL specific

WHERE

NULL

- `NULL` is not a value (it's the absence of a value)
 - to check null values, we use `IS NULL` or `IS NOT NULL`
 - `= NULL` will not work

LIKE

- `LIKE` allows for string wildcards
- `%` specifies the wildcard placement
 - `country_name LIKE 'and%'`
 - Andorra
 - `country_name LIKE '%and'`
 - Finland, Iceland ...more
 - `country_name LIKE '%and%'`
 - all of the above, *plus* Antigua and Barbuda, Netherlands, Rwanda ...more!
 - `country_name LIKE '%an%d%'`
 - Canada ...surely more!

WHERE

(WHERE live coding)

Building Queries:

Fundamental Three Commands

Two More Commands

Putting Things Together with JOIN

Two More Commands

CASE

DISTINCT

Two More Commands

CASE

DISTINCT

CASE

- `CASE` statements allow us to introduce conditional logic into our `SELECT` statements
- They are generally similar to `if` or `if else` statements in python, R, and other languages
 - When a condition is introduced, we check whether it evaluates to TRUE
 - If it is true, we proceed with a desired command, calculation, value, etc
 - If it is not true, we move to the next condition
 - If it is true, we proceed with another desired command, calculation, value, etc
 - ...all the way until we run out of conditions
 - For all FALSE conditions, we can use an `ELSE` statement if we want to
- The results of a `CASE` statement will be a new column
- Best practice is to name the new column using `AS new_column_name`

```
CASE
  WHEN [something is true]
    THEN [value or calculation]
  WHEN [something else is true]
    THEN [value or calculation]
  ELSE [value or calculation]
END
```

CASE

(CASE live coding)

Two More Commands

CASE

DISTINCT

DISTINCT

- Not all queries will result in unique rows (i.e. duplicates are present)
 - **Can we think of why this is? Write your thoughts in the etherpad!**
- `DISTINCT` has two possible spots within a query:
 - One comes immediately after `SELECT`, before column names are specified
 - e.g. `SELECT DISTINCT songs, albums, artists...`
 - This `DISTINCT` will govern the entire query
 - The other comes within aggregation (we'll get to this later)
 - e.g. `COUNT(DISTINCT products)`
 - This `DISTINCT` will only affect this specific aggregation

DISTINCT

(`DISTINCT` live coding)

Building Queries:

Fundamental Three Commands

Two More Commands

Putting Things Together with JOIN

Putting Things Together with JOIN

INNER JOIN

LEFT (OUTER) JOIN

RIGHT (OUTER) JOIN

FULL OUTER JOIN

Multiple Table Joins

Putting Things Together with JOIN

- Joins are used to combine data stored in different tables into a single result set
- Joins are the "Cartesian product" of two tables with *conditional selection(s)* of specific rows
 - A Cartesian product combines all possible row values with another
 - An easy example is a deck of cards:
combining four suits: {♠, ♥, ♦, ♣}
with thirteen ranks: {A, K, Q, J, 10, 9, 8, 7, 6, 5, 4, 3, 2}
produces 52 cards ($4 * 13$)
 - To create a Cartesian Product in SQL we use `CROSS JOIN` (rare, but not unheard of)

Putting Things Together with JOIN

- Joins require relationships (with one exception, `CROSS JOIN`) between tables
- Different joins create different results
 - Join names specify which conditional selection is desired
- There are three join types in SQL but different joining criteria can further limit results
- The most permitting join is a `FULL OUTER JOIN` and the least permitting is an `INNER JOIN`
 - Let's explore what this means by looking at each of them

Putting Things Together with JOIN

JOIN Syntax

Syntax for a join is as follows:

```
SELECT [columns]
FROM [left table]
JOIN [right table]
ON [left table.matching column] = [right table.matching column]
```

A couple of notes:

- You will need to specify which join type is desired:
 - e.g. `INNER JOIN`
- Matching columns do not need to have the same name, just the same value
 - e.g. `ON table1.LetterGrade = table2.Alphabet` will work because A=A, B=B, C=C, etc
- You can specify more than one column to be joined
 - e.g. `ON table1.FirstName = table2.FirstName AND table1.LastName = table2.LastName`

Putting Things Together with JOIN

INNER JOIN

LEFT (OUTER) JOIN

RIGHT (OUTER) JOIN

FULL OUTER JOIN

Multiple Table Joins

INNER JOIN

- **INNER JOIN** filters both tables to rows present in both tables
- **INNER JOIN** does not produce **NULL** values
- **INNER JOIN** is the "default" join
 - i.e. queries do not need to specify "INNER", though it's good practice to write INNER

Inner Join

Only rows from the "right table" and "left table" where values in the specified fields have matches in both tables

Colour	Quantity		Colour	Quantity
Pink	1	---	Pink	1
Blue	1		Teal	3
Green	2	---	Green	2
Yellow	2		Gold	2
Black	6	---	Black	6
Orange	3	---	Orange	3
Red	1		Crimson	2
Purple	1		Purple	2

Inner Join Result Set

	Colour	Quantity	Colour	Quantity
1	Pink	1	Pink	1
2	Green	2	Green	2
3	Black	6	Black	6
4	Orange	3	Orange	3

Image: Teate, Chapter 5 (+annotations by me)

INNER JOIN

A quick note on table aliasing:

- It is very common practice to alias table names
 - It makes join criteria much more concise
 - It simplifies `SELECT` statements when column names are the same
 - This is a common error: *"ambiguous column name"*
 - SQL requires you to specify *which* table you are returning the result from
- Generally, tables are aliased with the first letter (or first few letters) of the table so they can be easily referenced
 - `product AS p`
 - `product_category AS pc`

INNER JOIN

(INNER JOIN live coding)

Putting Things Together with JOIN

INNER JOIN

LEFT (OUTER) JOIN

RIGHT (OUTER) JOIN

FULL (OUTER) JOIN

Multiple Table Joins

LEFT (OUTER) JOIN

- `LEFT JOIN` filters the "right" table to rows present in the "left" table
- `LEFT JOIN` will most often produce `NULL` values
- The "OUTER" in `LEFT OUTER JOIN` is optional
 - Generally, OUTER seems to be excluded, but both are correct
- `LEFT` *is not* optional; there is no "OUTER JOIN"

Left Join

All rows from the "left table", and only rows from the "right table" with matching values in the specified fields

Colour	Quantity		Colour	Quantity
Pink	1	---	Pink	1
Blue	1		Teal	3
Green	2	---	Green	2
Yellow	2		Gold	2
Black	6	---	Black	6
Orange	3	---	Orange	3
Red	1		Crimson	2
Purple	1		Purple	2

Left Join Result Set

	Colour	Quantity	Colour	Quantity
1	Pink	1	Pink	1
2	Blue	1	<i>NULL</i>	<i>NULL</i>
3	Green	2	Green	2
4	Yellow	2	<i>NULL</i>	<i>NULL</i>
5	Black	6	Black	6
6	Orange	3	Orange	3
7	Red	1	<i>NULL</i>	<i>NULL</i>
8	Purple	1	<i>NULL</i>	<i>NULL</i>

LEFT (OUTER) JOIN

(LEFT JOIN live coding)

Putting Things Together with JOIN

INNER JOIN

LEFT (OUTER) JOIN

RIGHT (OUTER) JOIN

FULL (OUTER) JOIN

Multiple Table Joins

RIGHT (OUTER) JOIN

- **RIGHT JOIN** filters the "left" table to rows present in the "right" table
- **RIGHT JOIN** will most often produce **NULL** values
- The "OUTER" in **RIGHT OUTER JOIN** is optional
 - Generally, OUTER seems to be excluded, but both are correct
- **RIGHT JOIN** is somewhat frowned upon, but sometimes they make sense
 - Often your query can be reorganized to use a **LEFT JOIN** instead

Right Join

All rows from the "right table", and only rows from the "left table" with matching values in the specified fields

Colour	Quantity		Colour	Quantity
Pink	1	---	Pink	1
Blue	1		Teal	3
Green	2	---	Green	2
Yellow	2		Gold	2
Black	6	---	Black	6
Orange	3	---	Orange	3
Red	1		Crimson	2
Purple	1		Purple	2

Right Join Result Set

	Colour	Quantity	Colour	Quantity
1	Pink	1	Pink	1
2	Green	2	Green	2
3	Black	6	Black	6
4	Orange	3	Orange	3
5	NULL	NULL	Teal	3
6	NULL	NULL	Gold	2
7	NULL	NULL	Crimson	2
8	NULL	NULL	Purple	2

Putting Things Together with JOIN

INNER JOIN

LEFT (OUTER) JOIN

RIGHT (OUTER) JOIN

FULL (OUTER) JOIN

Multiple Table Joins

FULL (OUTER) JOIN

- `FULL OUTER JOIN` does not filter either "left" or "right" table
- Expect `NULL` values to be produced from a `FULL OUTER JOIN`
- My experience has been to write `FULL OUTER JOIN` rather than `FULL JOIN` but this is personal preference

Full Outer Join Result Set

	Colour	Quantity	Colour	Quantity
1	Pink	1	Pink	1
2	Blue	1	<i>NULL</i>	<i>NULL</i>
3	Green	2	Green	2
4	Yellow	2	<i>NULL</i>	<i>NULL</i>
5	Black	6	Black	6
6	Orange	3	Orange	3
7	Red	1	<i>NULL</i>	<i>NULL</i>
8	Purple	1	<i>NULL</i>	<i>NULL</i>
9	<i>NULL</i>	<i>NULL</i>	Teal	3
10	<i>NULL</i>	<i>NULL</i>	Gold	2
11	<i>NULL</i>	<i>NULL</i>	Crimson	2
12	<i>NULL</i>	<i>NULL</i>	Purple	2

Filtering a FULL (OUTER) JOIN

- All OUTER JOIN syntax can be filtered to exclude the *matching* criteria
 - Often called an ANTI JOIN, i.e. what's *not* in the other table

```
SELECT *  
FROM table_1  
{LEFT | RIGHT | FULL} OUTER JOIN table_2  
ON table_1.key = table_2.key  
WHERE {table_1.key IS NULL | table_2.key IS NULL |  
      table_1.key IS NULL OR table_2.key IS NULL}
```

Anti Join Result Set

	Colour	Quantity	Colour	Quantity
1	Blue	1	<i>NULL</i>	<i>NULL</i>
2	Yellow	2	<i>NULL</i>	<i>NULL</i>
3	Red	1	<i>NULL</i>	<i>NULL</i>
4	Purple	1	<i>NULL</i>	<i>NULL</i>
5	<i>NULL</i>	<i>NULL</i>	Teal	3
6	<i>NULL</i>	<i>NULL</i>	Gold	2
7	<i>NULL</i>	<i>NULL</i>	Crimson	2
8	<i>NULL</i>	<i>NULL</i>	Purple	2

Putting Things Together with JOIN

INNER JOIN

LEFT (OUTER) JOIN

RIGHT (OUTER) JOIN

FULL OUTER JOIN

Multiple Table Joins

Multiple Table Joins

- More than one table can be joined at a time

```
SELECT *  
FROM table_1  
{INNER | LEFT | FULL JOIN} table_2  
  ON table_1.key = table_2.key  
{INNER | LEFT | FULL JOIN} table_3  
  ON {table_1 | table_2}.key = table_3.key  
{INNER | LEFT | FULL JOIN} table_n  
  ON {table_1 | table_2 | table_3}.key = table_n.key
```

- The order and type of joins will have significant effect on the final result set
- It's important to determine which table should be the `FROM` table
- Sometimes you have to experiment a bit to get things right
- **Can you imagine scenarios based on your knowledge of different `JOIN` types that result in significantly different outputs?**

Multiple Table Joins

(Multiple Table Joins live coding)

